Answer Key

Section I

Select the best answer for the following questions (2 points each):

1. What is the upper warm layer of lakes known as?
    a. Hypolimnion
    b. Thermocline
    c. Pycnocline
    d. Epilimnion

2. In what region are lakes most common in?
    a. Former deserts
    b. Recently glaciated
    c. Forested areas
    d. Plains

3. Of the total water supply on Earth, how much is stored as groundwater in aquifers?
    a. Less than 1%
    b. About 3%
    c. About 5%
    d. About 12%

4. What is the layer of subsoil that is permanently frozen?
    a. Permafrost
    b. Active layer
    c. Talik
    d. Ice wedge

5. What is it called when water accumulates above a water table in an unsaturated zone?
    a. Settled
    b. Standing
    c. Perched
    d. Displaced

6. What mineral does hard water contain large amounts of?
    a. Calcium
    b. Sodium
    c. Fluoride
    d. Iron

7. Which of the following have the highest porosity?
    a. Granite
    b. Clay
    c. Shale
    d. Sand

8. Which karst feature forms on the ground of caves?
    a. Stacks
    b. Cave towers
    c. Stalactites
    d. Stalagmites

9. A low-lying mound with permanently frozen peat at the core is known as a(n) ___
    - **a. Palsa**
    - b. Pingo
    - c. Ice wedge
    - d. Kettle

10. Particles of sediment that roll and slide along the bottom of a stream are called:
    - **a. Bedload**
    - b. Water load
    - c. Suspended load
    - d. Displaced load

11. What are the curves and bends of a channel called?
    - a. Fans
    - b. Dendrites
    - **c. Meanders**
    - d. Braids

12. Which of the following are oxbow lakes most associated with?
    - **a. Meandering channels**
    - b. Anastomosing channels
    - c. Braided channels
    - d. Straight channels

13. Accumulations of sediment formed by the movement of water on the sides of a flood plain are called ____
    - **a. Natural levees**
    - b. Deltas
    - c. Alluvial fans
    - d. Backswamp

14. In the profile of a typical river, how does the gradient change from head to mouth?
    - a. Increases
    - **b. Decreases**
    - c. Does not change
    - d. Varies constantly

15. How far can large rivers flow into the ocean and maintain freshwater before mixing with the ocean?
    - a. A few meters
    - b. Tens of meters
    - c. A few kilometers
    - **d. Many kilometers**

16. Which of the following acids that natural water contains makes it easy for groundwater to dissolve limestone?
    - a. Acetic acid
    - b. Nitric acid
    - **c. Carbonic acid**
    - d. Hydrochloric acid

17. Alluvial fans are made of sediments deposited at _____
    a. The mouth of a river
    b. The banks of a river
    c. The intersection between the river and ocean
    d. The inside of a meander loop

18. The transition of water directly from glacial ice to liquid is known as:
    a. Transpiration
    b. Sublimation
    c. Deposition
    d. Condensation

19. Rivers that lose water as they flow downstream are
    a. Influent stream
    b. Effluent stream
    c. Discharging stream
    d. Recharging stream

20. Speleothems in karst caves are most commonly made of
    a. Chlorite
    b. Calcite
    c. Magnesium
    d. Sulfur

21. Globally, how much of the freshwater supply is used for agriculture?
    a. Under 5%
    b. Around 15%
    c. Around 50%
    d. Over 70%

22. The following are associated with periglacial regions except:
    a. Pingos
    b. Permafrost
    c. Ice sheets
    d. Ice wedges

23. If a stream drops 20 meters in 10 kilometers, what is its gradient?
    a. 2 meters per kilometer
    b. 2 kilometers per meter
    c. 0.5 meter per kilometer
    d. 0.5 kilometer per meter

24. Which of the following occurs when a river flows through a delta to the sea?
    a. Discharge of the river increases
    b. Sediment size of the river increases
    c. Total sediment load of the river decreases
    d. Height of the river increases

25. A stream's ability to erode and transport material mainly depends on its:
    a. Length

b. Velocity
c. Sinuosity
d. Gradient

26. What are artificial levees meant to control?
    a. Meandering
    b. Flooding
    c. Erosion
    d. Velocity

27. Streams that undergo downcutting cause __-shaped valleys.
    a. U
    b. V
    c. W
    d. Y

28. What is the ultimate base level?
    a. The highest point in a stream
    b. The same level of the nearest lake is reached
    c. Sea level
    d. The lowest level of an aquifer

Select all that apply for the following questions (3 points each):

29. Which of the following are shortcomings of dams?
    - **a. Expensive to create**
    - b. Cause significantly increased flooding
    - **c. Destroy existing ecosystems**
    - **d. Increase wastewater**
    - e. Reduce drinking water

30. Which of the following are qualities of the best groundwater aquifers?
    - a. Low permeability
    - b. Low porosity
    - **c. High permeability**
    - **d. High porosity**
    - e. None of the qualities above are significant

31. What conditions lead to a parallel drainage pattern?
    - **a. Area with steep slopes**
    - b. Volcanoes
    - **c. Parallel elongated regions of rock**
    - d. System of fractures and faults
    - e. Above flat-lying sedimentary rock

32. Which of the following describes a graded stream?
    - **a. Balance of erosional and depositional processes**
    - b. Balance of coarse and fine sediments
    - c. Constant velocity and discharge
    - d. Lack of erosion and deposition
    - **e. A system in equilibrium**

33. Which are true of playa lakes?
    - **a. Formed from filling in dry lake beds**
    - **b. Found only temporarily after rainfall**
    - c. Permanent source of water
    - **d. Found most commonly in desert regions**
    - e. Receives less than 100 inches of rainfall a year

34. Which contributes to the formation of stream valleys?
    - a. Porosity
    - b. Capacity
    - **c. Weathering**
    - **d. Running water**
    - **e. Mass wasting**

Matching

35. j
36. l
37. m
38. d
39. n

40. c
41. k
42. f
43. a
44. h
45. g
46. e
47. i
48. b

Section II

Question I

1. —
    a. Cenote
    b. Exposes groundwater at bottom
    c. Mexico (Yucatan)
2. Limestone
    a. Dolomite/dolostone, gypsum (possibly others)
    b. Dissolution of the calcium carbonate rich rocks as surface water/groundwater enters the cracks, fractures, and holes in the bedrock
    c. $CaCO_3 + H_2O + CO_2 \rightarrow Ca^{2+} + 2\ HCO_3^-$
3. True
    a. Karst aquifers (especially shallow ones) are highly susceptible to contamination. Karst features such as sinking streams and sinkholes form a rapid route with reduced filtration as water moves from the surface to the aquifers.
4. South China Karst

Question II

1. (**0.5 points** for each correct identification) P = precipitation, E = evaporation, ΔS = change in storage, Q = runoff
    a. Groundwater/bedrock, soil moisture, snow/ice, lakes/reservoirs, etc. (storage source of water) (**1 point** for each source, max **2**)
2. Throughfall, stemflow and interception loss
    a. (**2 points** for each) Throughfall is the water that either directly falls to the ground (through gaps) or indirectly (dripping off the canopy). Stemflow is the water intercepted by stems/branches and flows down the trunk into the soil. Interception is the water that does not reach the soil but is intercepted; interception loss occurs as a form of evaporation when the water is on the canopy prior to throughfall or stemflow.

Question III

1. Meandering
2. —
    a. Oxbow lake
    b. Cutbank
    c. Cutoff
    d. Point bar
    e. Back swamp
3. Oxbow lakes are formed when a river is cut off from a meandering channel, to create a free standing body of water. Sediment builds up on the inside of the curve, becoming more curvy, and ultimately forming a cutoff. The original curve eventually separates to form the lake.
4. As glaciers move back and forth, they create deep basins that eventually become lakes as the ice melts.
    a. Great Lakes, etc.
5. —

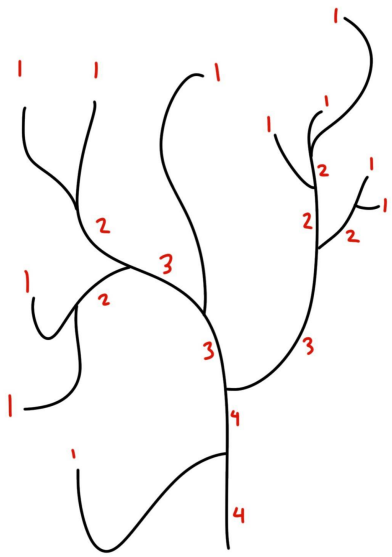
    a. Fall
    b. Spring
    c. Winter
    d. Summer
    e. Due to spring turnover. This is mixing of the entire lake to create an even temperature as the surface water warms in spring and equalizes the temperature in the lake layers.
6. —
    a. Epilimnion
    b. Metalimnion
    c. Thermocline
    d. Hypolimnion

Question IV

1. More rounded indicates that the rocks have been exposed to longer transportation and/or strong winds and wave activity.
2. Breccia are angular clasts; conglomerate are rounded clasts.
3. Direct
4. Graded bedding
    a. Indicates a strong, sometimes sudden fluid flow that deposits heavy, coarse sediments first, with finer ones layered above.

Question V

1. Radial
2. Bird's foot
    a. Occurs when river flow is stronger and wave flow is weaker. Sediment deposits occur at the mouth of the river and divide into several distributaries, forming the "bird's foot" shape.
    b. Mississippi River
3. (1 point for each correct)
    a. 2
    b. 2
    c. 3
    d. 4
4. River flows with harder rock on top and softer rock below (**1 point**), causing water to wear away at the soft work to form a plunge pool (**1 point**). Erosion continues until a gorge is left behind (**1 point**).



Question VI

1. A flood has a 5% chance of happening each year. When examining data over a long period of time, there will be approximately one flood every 20 years.

2. Southwest/Midwest (**1 point**). This region is more arid (**1 point**) and the streams stop flowing without as much precipitation/runoff to source the streams due to the lack of groundwater storage (**1 point**).
3. Using an Acoustic Doppler Current Profiler (ADCP), we determine water velocity by sending a sound pulse out into the water and determining the change in frequency of the reflected pulse (**1 point**). The water velocity is used to calculate the stream flow of the water, cross sectional area x velocity (**1 point**).
    a. 0.8 for rocky, loose, or coarse bottom streams, 0.9 for muddy, sandy, or smooth bottom streams (**2 points**). The stream velocity we calculate is the surface velocity, and the correction factor accounts for the fact that water at the bottom travels slower due to resistance from the rocks, etc. Rocks and coarser-bottom streams have more resistance and overall slower stream velocity (**2 points**).
4. (**up to 2 points for correct work/partial credit, 1 point for correct answer**)

    $r = 0.1$ m $h = 0.05$ m

    $\theta = 2\arccos((0.1-0.05)/0.1) = 2.09$ rad

    $A = 0.006$ m^2 (area of flow)

    $P = 0.209$ m (wetted perimeter)

    $R_h = 0.029$ m (hydraulic radius)

    a. The water has higher flow velocity.

Question VII

1. —
    a. Zone of aeration
    b. Capillary fringe
    c. Water table
    d. Zone of saturation
2. Gravel, sand, limestone, etc.
    a. The material must be highly porous and permeable so that it permits water to flow through.
3. False.
    a. The soil filters out the large impurities but many dissolved minerals and chemicals that are too small to see remain.
4. (Varied answers - 1 point for valid scenario, up to 2 points for explanation) Can be used in the oil/petroleum industry to calculate the flow in reservoirs of porous materials.
    a. Darcy's Law can generally only be used when the flow is laminar, not turbulent — i.e., water flows in parallel layers without disruptions in between.

Question VIII

1. Fish ladder; gives migrating fish a path to move around the dam that would otherwise block their path.

2. Bacteria and other microorganisms use up oxygen while breaking down the sewage, causing oxygen depletion in certain areas. Other organisms that require oxygen such as fish will be unable to survive.
3. Varied answers - any that make sense (ex. don’t overuse pesticides, don’t litter in rivers, etc.)